

4. OPEN QUESTIONS

Question 1. Clearly, the first question we are interested in is whether there are inner functions Θ , other than the meromorphic inner functions $e^{i\tau(\cdot)}$, such that the associated Toeplitz operator $T_{\overline{\Theta}}: H_{\Theta}^1 \rightarrow H^1(\mathbb{C}^+)$ is bounded, or equivalently, for which inner functions Θ the equality

$$H_{\Theta}^1 = K_{\Theta}^1 \oplus \Theta H^1$$

holds. Moreover, from Theorem 1.3 we know that, if Θ is an inner function, $T_{\overline{\Theta}}: H_{\Theta}^1 \rightarrow H^1(\mathbb{C}^+)$ is unbounded if and only if

$$K_{\Theta}^1 \oplus \Theta H^1 \subsetneq H_{\Theta}^1,$$

that is, if and only if the quotient $H_{\Theta}^1/(K_{\Theta}^1 \oplus \Theta H^1)$ is non-trivial. When non-trivial, it would be interesting to describe it.

Question 2. The second question delves deeper into the subject and it is related to another interesting problem. In analogy with the holomorphic case, we introduce a subspace of the real variable Hardy space $H^1(\mathbb{R})$. For $b \in \text{BMO}(\mathbb{R})$ we consider the pre-annihilator of $\{b\}$, that is,

$$(4.1) \quad H_b^1(\mathbb{R}) := \{\phi \in H^1(\mathbb{R}) \text{ such that } \langle \phi, b \rangle = 0\}.$$

Observe that, if $b = \Theta \in H^\infty(\mathbb{C}^+)$

$$H_{\Theta}^1(\mathbb{R}) = \overline{H^1(\mathbb{C}^+)}|_{\mathbb{R}} \oplus (H_{\Theta}^1)|_{\mathbb{R}}.$$

Following the ideas in Theorem 1.3, we reformulate the boundedness of $T_{\overline{\Theta}}$ in terms of $[\overline{\Theta}, H]$.

Proposition 4.1. *Let Θ be an inner function. Then, the following are equivalent:*

- (a) *the Toeplitz operator $T_{\overline{\Theta}}: H_{\Theta}^1 \rightarrow H^1(\mathbb{C}^+)$ is bounded;*
- (b) *the commutator $[\overline{\Theta}, H]: H_{\Theta}^1(\mathbb{R}) \rightarrow L^1(\mathbb{R})$ is bounded.*
- (c) *the multiplication operator $M_{\overline{\Theta}}: H_{\Theta}^1(\mathbb{R}) \rightarrow H^1(\mathbb{R})$ is bounded.*

Proof. We note that for every $H_{\Theta}^1(\mathbb{R}) \ni f = f_1 + f_2$, with $f_1 \in \overline{H^1(\mathbb{C}^+)}|_{\mathbb{R}}$ and $f_2 \in H_{\Theta}^1$, by using (1.2) we have that

$$[\overline{\Theta}, H](f_1) = \overline{\Theta}H(f_1) - H(\overline{\Theta}f_1) = \overline{\Theta}(-if_1) + i\overline{\Theta}f_1 = 0.$$

Moreover,

$$[\overline{\Theta}, H](f_2) = \overline{\Theta}H(f_2) - H(\overline{\Theta}f_2) = -i\overline{\Theta}f_2 + iP_+(\overline{\Theta}f_2) - iP_-(\overline{\Theta}f_2) = -2iH_{\overline{\Theta}}(f_2).$$

Consequently, $[\overline{\Theta}, H]$ is bounded if and only if the Hankel operator $H_{\overline{\Theta}}$ is bounded from H_{Θ}^1 to $\overline{H^1(\mathbb{C}^+)}|_{\mathbb{R}}$, which is equivalent to the boundedness of $T_{\overline{\Theta}}: H_{\Theta}^1 \rightarrow H^1(\mathbb{C}^+)$, by Theorem 1.3. Thus, (a) and (b) are equivalent.

Next, suppose (a) holds. Then we have that if $f \in H_{\Theta}^1$, then $(\overline{\Theta}f)|_{\mathbb{R}} \in H^1(\mathbb{R})$. Indeed, we wish to show that $(\overline{\Theta}f)|_{\mathbb{R}}$ and $H((\overline{\Theta}f)|_{\mathbb{R}}) \in L^1(\mathbb{R})$. Clearly $\overline{\Theta}f \in L^1(\mathbb{R})$, and

$$H(\overline{\Theta}f) = -i(2P_+ - \text{Id})(\overline{\Theta}f) = -2iT_{\Theta}f + i\overline{\Theta}f \in L^1(\mathbb{R}),$$

since (a) holds. Therefore, $(\overline{\Theta}f)|_{\mathbb{R}} \in H^1(\mathbb{R})$, and

$$\|\overline{\Theta}f\|_{H^1(\mathbb{R})} \leq 2\|T_{\Theta}f\|_{H^1(\mathbb{C}^+)} + \|f\|_{H^1(\mathbb{C}^+)} \leq C\|f\|_{H^1(\mathbb{C}^+)},$$